Comparing fixed-OH and joint methane–OH inversions

Reading note on Feng et al. (2023)

Methane emissions are predominantly responsible for record-breaking atmospheric methane growth rates in 2020 and 2021. Atmospheric Chemistry and Physics 23, 4863–4880. doi:10.5194/acp-23-4863-2023. Appendices A–C are included in the published article; no separate supplement is listed.

Method. The ensemble Kalman filter jointly assimilates Leicester GOSAT v9 CH_4:CO_2 column ratios and surface measurements of both gases. It therefore avoids assigning the proxy ratio to methane using prescribed CO_2 alone. GEOS-Chem uses MERRA2 meteorology at $2^\circ \times 2.5^\circ$. Monthly emission factors cover 476 land and 11 ocean regions, with a four-month assimilation window. Prior methane flux uncertainty is 60%, with 300-km spatial and one-month temporal error correlation. Alongside a fixed-OH control, a joint inversion for 2018–2021 optimizes six monthly zonal OH factors with 3% prior uncertainty. Polar OH is not adjusted. Wetlands and rice follow WetCHARTs; other anthropogenic priors use EDGAR v4.3.2 for 2012.

Findings. Table 1 gives fixed-OH emissions of 583.7, 610.7, and 604.5 Tg yr^{-1} for 2019, 2020, and 2021, respectively. The joint inversion gives 585.3, 603.2, and 603.7 Tg yr^{-1}, so its 2019-relative anomalies are 17.9 and 18.4 Tg yr^{-1}. Published OH changes relative to the prior are +0.91, −0.52, and +0.62%; the text reports changes of −1.43 and −0.29 percentage points relative to 2019. The 34% attribution in 2020 is the reduction of the inferred emission anomaly from 27.0 to 17.9 Tg yr^{-1} when OH is optimized. This is a comparison of two inversions, not a direct evaluation of $-m_0\Delta k_{\mathrm{OH}}$. Their posterior regional flux–OH error correlations are generally below 0.05 in magnitude.

Eastern Africa contributes about 14 Tg yr^{-1} in the fixed-OH 2020 comparison. China and India decrease by about 6 and 2 Tg yr^{-1}. Tropical South America and temperate North America matter more in 2021. Regional water-storage and emission anomalies correlate, but the association alone does not distinguish microbial sectors.

Comparison decisions. Appendix B explicitly distinguishes growth calculated between calendar-year endpoints from differences between annual means: the latter shift the effective timing by roughly half a year. We retain this distinction and use each inversion's own 2019 emissions. Annual posterior errors of roughly 11–13 Tg yr^{-1} must not be treated as uncertainty of year-to-year differences without the cross-year covariance. Table 1 can be transcribed exactly; no gridded posterior download is linked. The 9.1 Tg yr^{-1} fixed-versus-joint difference in 2020 is retained as a sensitivity diagnostic and is not labeled a measured OH forcing. TCCON validation and the sensitivity experiments support the inferred large-scale flux patterns, conditional on the assumed OH parameterization.